David Fisac

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Employment

Postdoctoral researcher at Laboratoire d'Analyse et de Mathématiques Appliquées (LAMA) for CNRS at Université Paris-Est Créteil, mentored by Federica Fanoni. Oct 2025 – Oct 2026

Postdoctoral researcher at Universitat Autònoma de Barcelona at the Universal Geometric Structures and Inequalities group. Feb 2025 – Sept 2025

Education

Université du Luxembourg - Universitat Autònoma de Barcelona, PhD in Mathematics, Feb 2021 – Jan 2025

Arcs, curves, and the surfaces they live on: e-print.
Under the co-supervision of Florent Balacheff and Hugo Parlier.

Imperial College London, MSc in Pure Mathematics *with Distinction*, Sept 2019 – Sept 2020

Conformal geodesics on Lorentzian conformal manifolds motivated by connectedness.
Under the supervision of Marie-Amélie Lawn.

Universitat Autònoma de Barcelona, Grau en Matemàtiques, Sept 2015 – June 2018

Menció en Matemàtica Fonamental.

Research Interests

I study hyperbolic, topological and Riemannian surfaces, of finite and infinite type. More specifically, arcs, curves, and laminations on them, their length spectra, and their intersection.

(Pre)Publications

Exact curve counting of given word length on the once-punctured torus, Fresh
with Filippo Baroni and Mingkun Liu,
arXiv:2609.01382.

Isospectrality for infinite-type hyperbolic surfaces with discrete length spectrum, Submitted
with Federica Fanoni,
arXiv:2602.19670.

Single self-intersection words on the once-punctured torus and their counting, Submitted
with Mingkun Liu,
arXiv:2404.09372.

A Basmajian-type inequality for Riemannian surfaces, Published
with Florent Balacheff,
Journal of Topology and Analysis 18 (04): 1199-1212 (2026),
arXiv:2311.03182.

Markov's conjecture on integral necklaces, Published
Bulletin of the London Mathematical Society, 57 (12): 4122-4131 (2025),

arXiv:2501.15550.

Organization

Curves, Hyperbolicity, And Teichmüller Spaces (CHATS), June 2026
with Yusen Long,
at Université Paris-Est Créteil.

Invited Talks

Recent developments in Teichmüler theory in Oberwolfach, Germany. Oct 2026
Géométrie hyperbolique et surfaces, in honor of Greg McShane at Institut Fourier, Grenoble, France. Sep 2026
Séminaire Géométrie et Topologie at Institut de Mathématiques de Jussieu, Paris, France. May 2026
Curve counting: what's next? (short) at the Hamilton Mathematics Institute, Dublin, Ireland. May 2026
Geometry and Spectra of Compact Riemann Surfaces (lightning) at EPFL, Lausanne, Switzerland. Apr 2026
Geometry and Topology seminar at the Université du Luxembourg, Luxembourg. Dec 2025
Geometry seminar at MPI MiS, Leipzig, Germany. Jun 2025
Combinatorics mittagsseminar at ETHZ, Zurich, Switzerland. May 2025
Perspectives on Markov Numbers (lightning) at BIRS, Banff, Canada. Jan 2025
Colloque de Géométrie at the Université Paris-Est Créteil, Paris, France. Nov 2024
2nd International AMS-UMI Joint Meeting, Palermo, Sicily. Jul 2024
Hengstberger Symposium: Geometry and dynamics of surfaces beyond finite-type, Heidelberg, Germany. Jul 2024
Surfaces, Manifolds and Related Topics (SMaRT) at the Université du Luxembourg, Luxembourg. Mar 2024
Seminari de Geometria at the UAB, Barcelona. Jan 2024
Geometry, Probability and their Synergies, Les Diablerets, Switzerland. Mar 2023
VI Congreso de Jóvenes Investigadores de la RSME, León, Spain. Feb 2023

References

Federica Fanoni, Directrice de Recherche, CNRS (Institut de Mathématiques de Jussieu), *federica.fanoni@gmail.com*.

Viveka Erlandsson, Associate Professor, CMAT (Universidad de la República in Uruguay), *verlandsson@cmat.edu.uy*.

Florent Balacheff, Professor Titular, Universitat Autònoma de Barcelona, *florent.balacheff@uab.cat*.

Hugo Parlier, Professeur ordinaire, Université de Fribourg, *hugo.parlier@unifr.ch*.

Teaching

Caculus for the Statistics BSc TA, Universitat Autònoma de Barcelona. 2025
Mathematics for Geology: Statistics TA, Universitat Autònoma de Barcelona. 2025
Math Forge (all-round tutoring for Mathematics BSc), University of Luxembourg. 2021-2023-2024
Linear Algebra I TA, University of Luxembourg. 2022

Languages

Catalan: native.
Spanish: native.
English: proficient.
Italian: proficient.
French: basic.

Programming

Languages: Python, C, Java.

Mathematical programming: Mathematica, SageMath, R.